

15
FEATURES

OCEANEMBED · SIH26066 · QUICK-ANSWER CARD

Every Feature, One Sentence Each

What to click, and what to say while it loads. Keep this beside you.

ERROR VS REAL FLOATS

0.9063 degC

INDEPENDENT PROFILES

963

BETTER THAN AVERAGE BY

23.8%

FEATURES ON THE RAIL

15

START EVERYTHING ONCE, THEN JUST CLICK THE RAIL

```
.venv/Scripts/python.exe -m streamlit run app/ui/main.py --server.port 8500
```

Set the **Date** once at the top and leave **Input** on **Satellite**. Those two apply to every feature below, so you never set them again.

SEE IT - What the system produces.

#	Feature	Do this	Say this
1	3-D Ocean	Click it, then drag to rotate; the volume is already built for the chosen date.	This is the whole output - 24,000 cells x 15 depths. The holes are the sea floor, not missing data.
2	Click a point	Click anywhere in the sea on the map; then click on land.	A float answers this only where it happens to be. We answer it everywhere - and on land we refuse instead of guessing.
3	Confidence	Click it, then switch View to uncertainty alone.	The model is least sure exactly where the ocean is most active - the Somali Current and the Arabian Sea eddies.
4	Transect	Type two end points (try 8 N 68 E to 20 N 88 E) to cut a vertical wall through the sea.	Structure is what oceanographers read - a tilting thermocline is obvious here and invisible on a map.
5	Sound	Click it and pick a Layer ; the sound-speed profile is computed from our temperature.	Sound speed follows temperature, so a temperature model is an acoustics model - this is what a navy actually acts on.

PROVE IT - Evidence that it is right.

#	Feature	Do this	Say this
6	Validation	Click it and walk down the per-depth table.	Measured against real floats it never saw. It beats the long-term average at 14 of 15 depths, and we label the one where it loses.
7	Hidden heatwaves	Click it; the detection is precomputed, so it appears instantly.	Five days above the seasonal 90th percentile at depth - a marine heatwave a satellite physically cannot see.
8	Cyclone heat	Click it, pick Show , and choose an October or November date.	Not surface warmth - the depth of the warm layer, which is what decides whether a storm intensifies.
9	Where to measure next	Click it; leave the sliders alone unless asked.	Where our doubt meets an energetic ocean. A suggestion of where to measure next, never an instruction.
10	Learn from a float	Click it; the experiment is precomputed.	A real float is fed back into the frozen model with no retraining, and the fix spreads to water in a similar state - not water nearby.

STRESS IT - Us attacking our own model.

#	Feature	Do this	Say this
11	Cyclone wake	Click it, pick a Storm , then switch Panel from the naive view to the passage-relative one.	Cooling at 11 of 12 track points, from a model never taught that cyclones cool the ocean. The naive view finds nothing - we show both.
12	Cloud cover	Click it; the sweep is precomputed.	We blanked the satellite input on purpose, like monsoon cloud. Past 20% the loss is real and steady.
13	Profile shape	Click it.	Error scores each depth alone. This asks whether the shape between depths survived - and the thermocline gradient does.
14	Physical profiles	Click it and read the before/after bars.	805 physically impossible columns become 0. A training penalty makes that rare; a projection makes it impossible.
15	What it can see	Click it and read the stacked bar down the depths.	It reads sea-surface temperature near the top and sea-surface height at the thermocline - untaught, and physically correct.

THE THREE WE BUILT THAT THE PUBLISHED WORK DOES NOT DO

Physical profiles - a hard physical guarantee where the field uses a soft penalty and therefore never reports a violation count. **What it can see** - a map of which satellite measurement informs which depth. **Learn from a float** - an observation fed back into the frozen model at run time. If asked what is novel, these three, and the system around them - not the reconstruction method, which is published prior art.

IF A FEATURE WILL NOT LOAD

Say so and move on - every panel refuses rather than inventing a number, and that refusal is the behaviour we designed. Do not restart the app mid-demo.

THE ONE NUMBER TO NEVER GET WRONG

0.9063 degC against 963 independent Argo profiles, under scoring protocol **seafloor_masked_v2**. Older documents may say 0.9078 - that was an earlier protocol on the same checkpoint, before below-seafloor comparisons were declined and the float depth axis was corrected. Quote the number above.